INTERNSHIP LEARNING SUMMARY

Introduction to Data Science, AI, and Key Techniques

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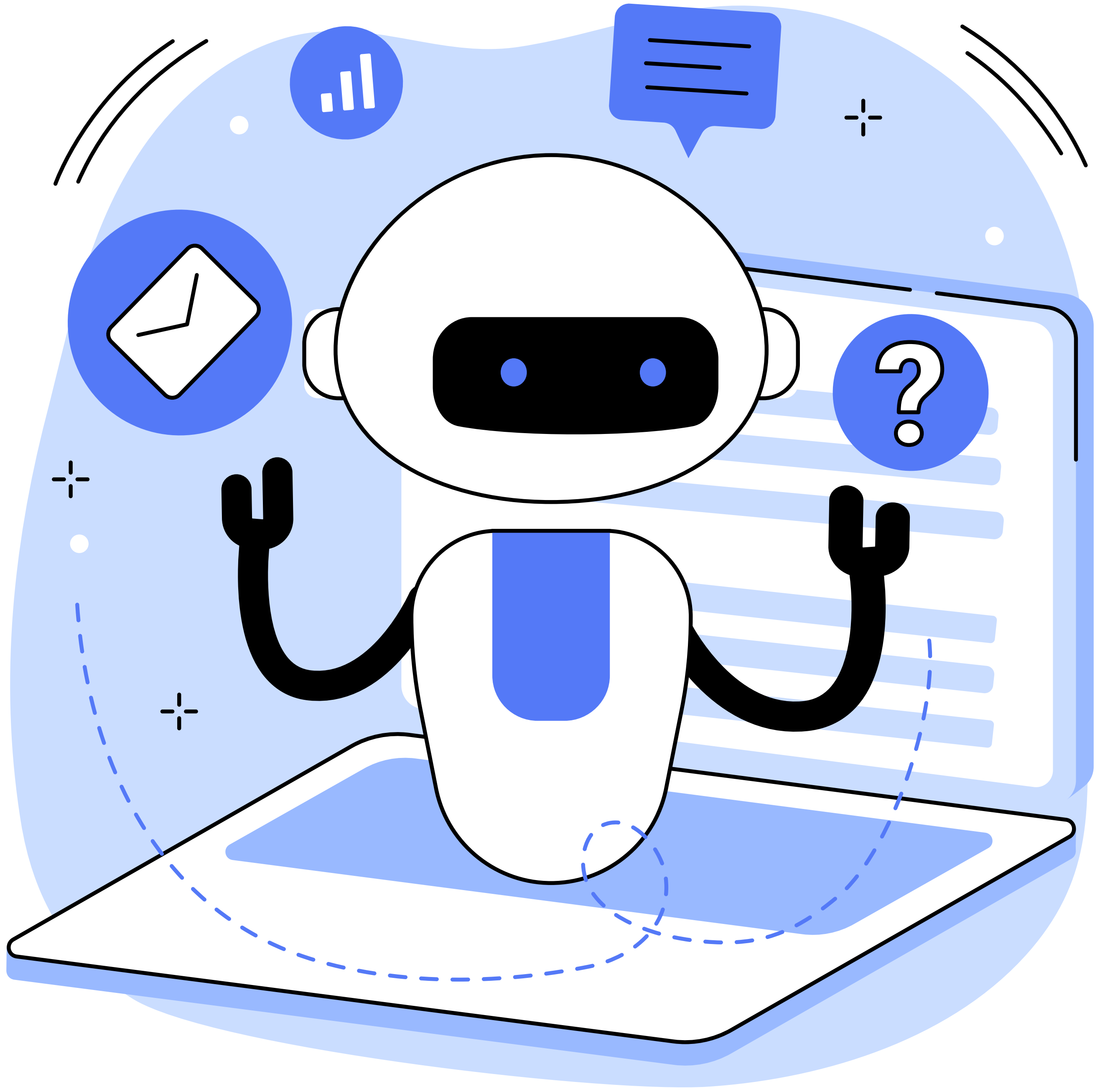
WHAT IS DATA SCIENCE?

- Interdisciplinary field combining domain knowledge, programming, and statistics
- **Goal:** extract useful insights from structured and unstructured data
- **Data Science Lifecycle:**
  - Define Problem → Collect Data → Wrangle Data → Explore Data → Model Data → Communicate Results
- **Real-world use:** recommendation systems, fraud detection, customer insights



WHAT IS ARTIFICIAL INTELLIGENCE (AI)?

- **AI:** Making machines mimic human intelligence
- **4 Perspectives:**

  Acting humanly, Thinking humanly

  Acting rationally, Thinking rationally

- **Types of AI:**
- Narrow AI (task-specific), General AI (multi-task), Super AI (theoretical)

- **AI subfields:** Machine Learning, Deep Learning, NLP, Robotics

WHAT IS BIG DATA?

- **Refers to data that is:**
  - **Volume:** Extremely large (TBs to PBs)
  - **Velocity:** Generated rapidly (real-time streams)
  - **Variety:** Structured, semi-structured, unstructured

- **Challenge:** Traditional tools can't handle it

- **Example:** 328 million TB generated daily (source: explodingtopics)

APPLICATIONS OF AI & DATA SCIENCE

Human Resource Management

Resume screening, onboarding automation

Finance

Risk analysis, fraud detection, trading strategies

Healthcare

Predictive diagnostics, drug discovery

Retail

Inventory management, dynamic pricing

Impact: Faster decisions, cost savings, reduced risks, competitive advantage

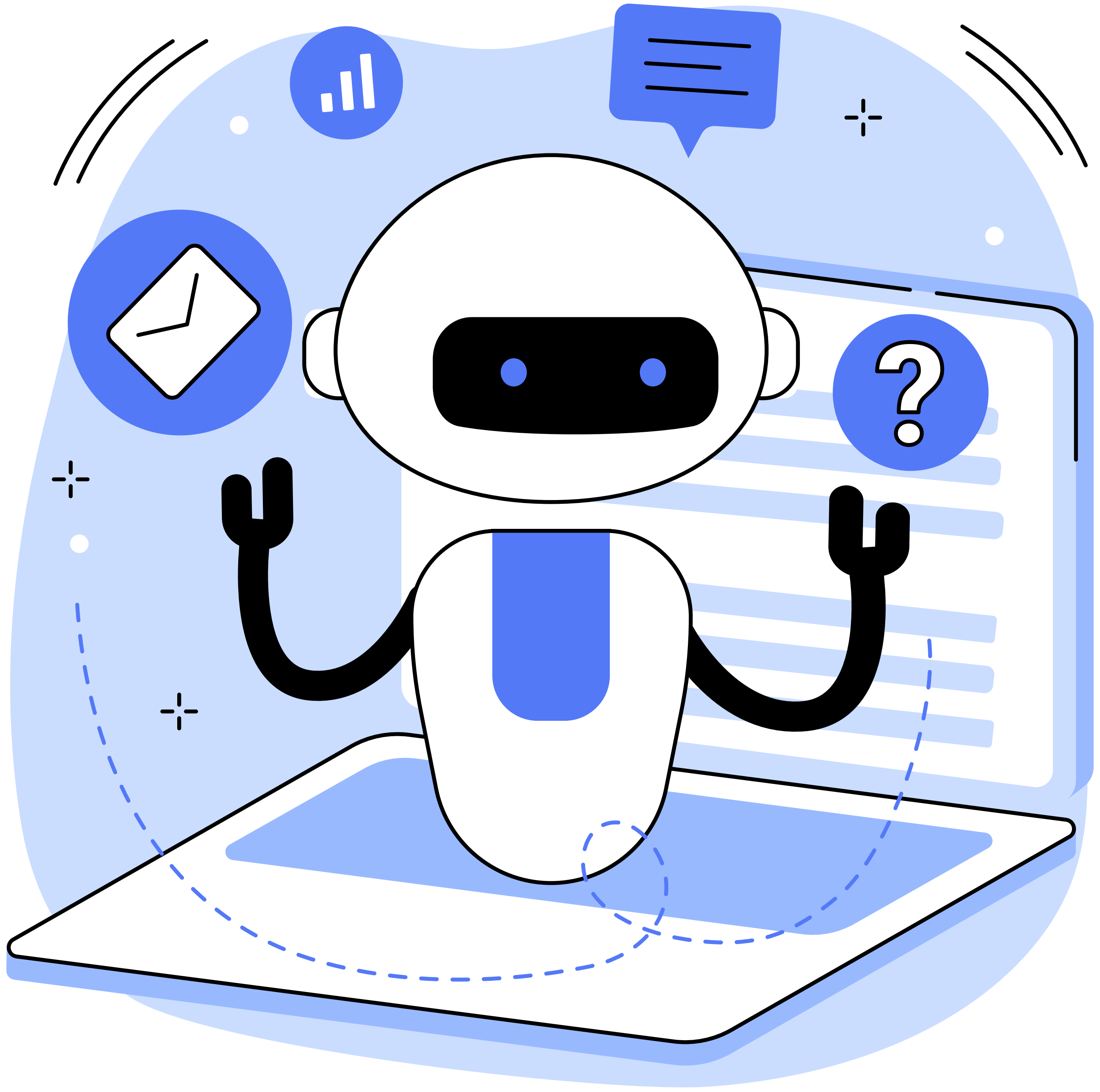
INTRODUCTION TO DATA WRANGLING

- Also called data munging or preprocessing
- **Purpose:** Clean messy, inconsistent data into usable form
- **Typical tasks:**
  - Detecting and handling nulls
  - Formatting columns (date, strings, numbers)
  - Removing duplicates and irrelevant data

Quote: “Clean data leads to clean results”

TOOLS AND LIBRARIES USED

- **pandas:** DataFrames, manipulation, cleaning
- **numpy:** Array operations, numerical computations
- **Methods learned:**
  .info(), .head(), .isnull(), .dropna()
  .fillna(), .astype(), .rename(), .duplicated()
- Feature scaling: Normalization, Standardization



DATA WRANGLING IN PRACTICE

Handling missing values

Drop or impute with mean/median/mode

Data type conversions

Strings to dates (pd.to_datetime())

Outlier detection

IQR method, Z-score

Encoding

One-hot, Label Encoding for categories

Impact: Faster decisions, cost savings, reduced risks, competitive advantage

INTRODUCTION TO DATA VISUALIZATION

- **Purpose:** Understand data structure, spot trends, communicate findings

- Tools explored:
  Matplotlib, Seaborn, pandas built-in plotting

- **Benefits:**
- Make data insights visually clear
- Aid in pattern recognition and storytelling

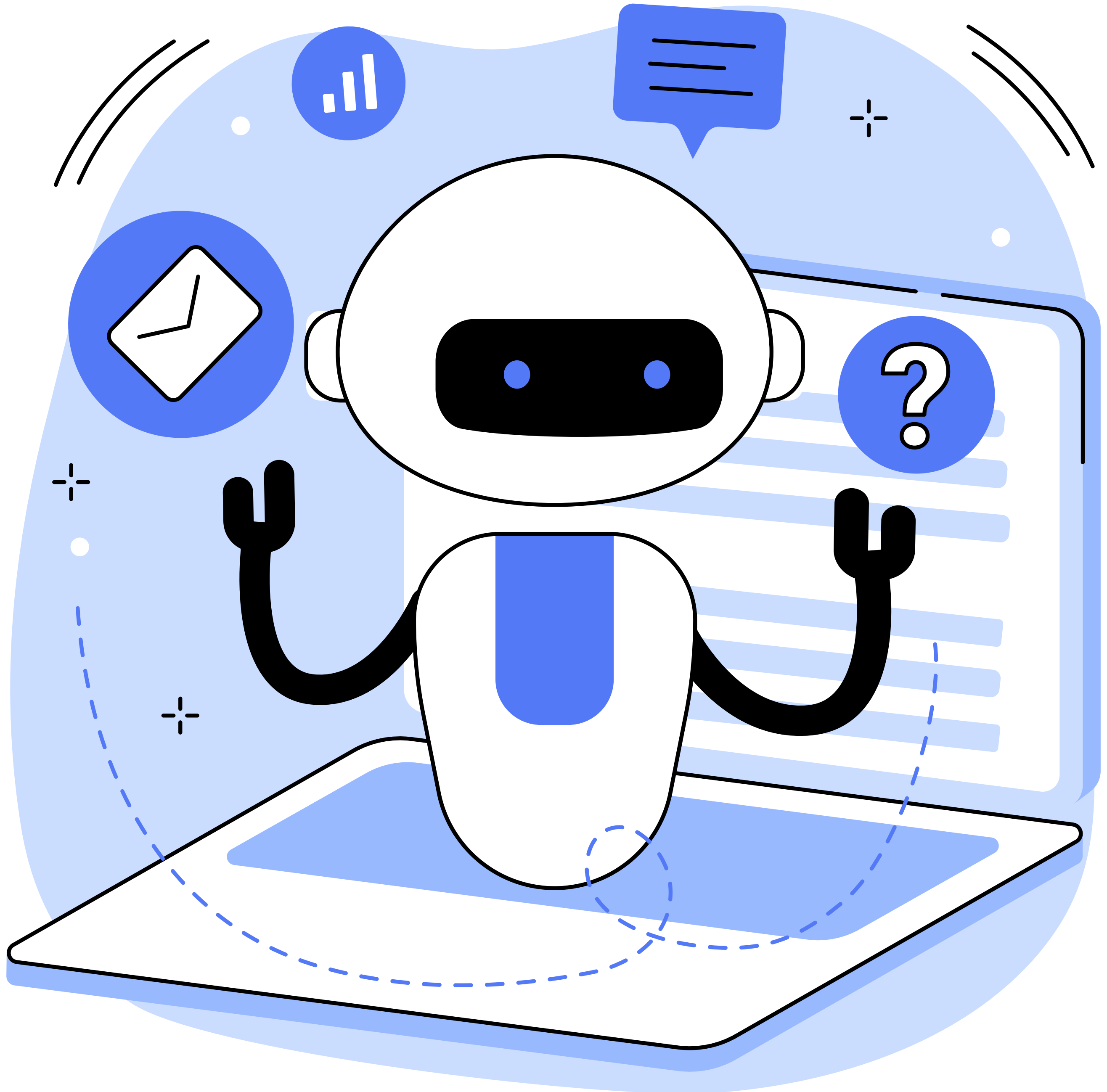
MATPLOTLIB BASICS

- **Two main approaches:**

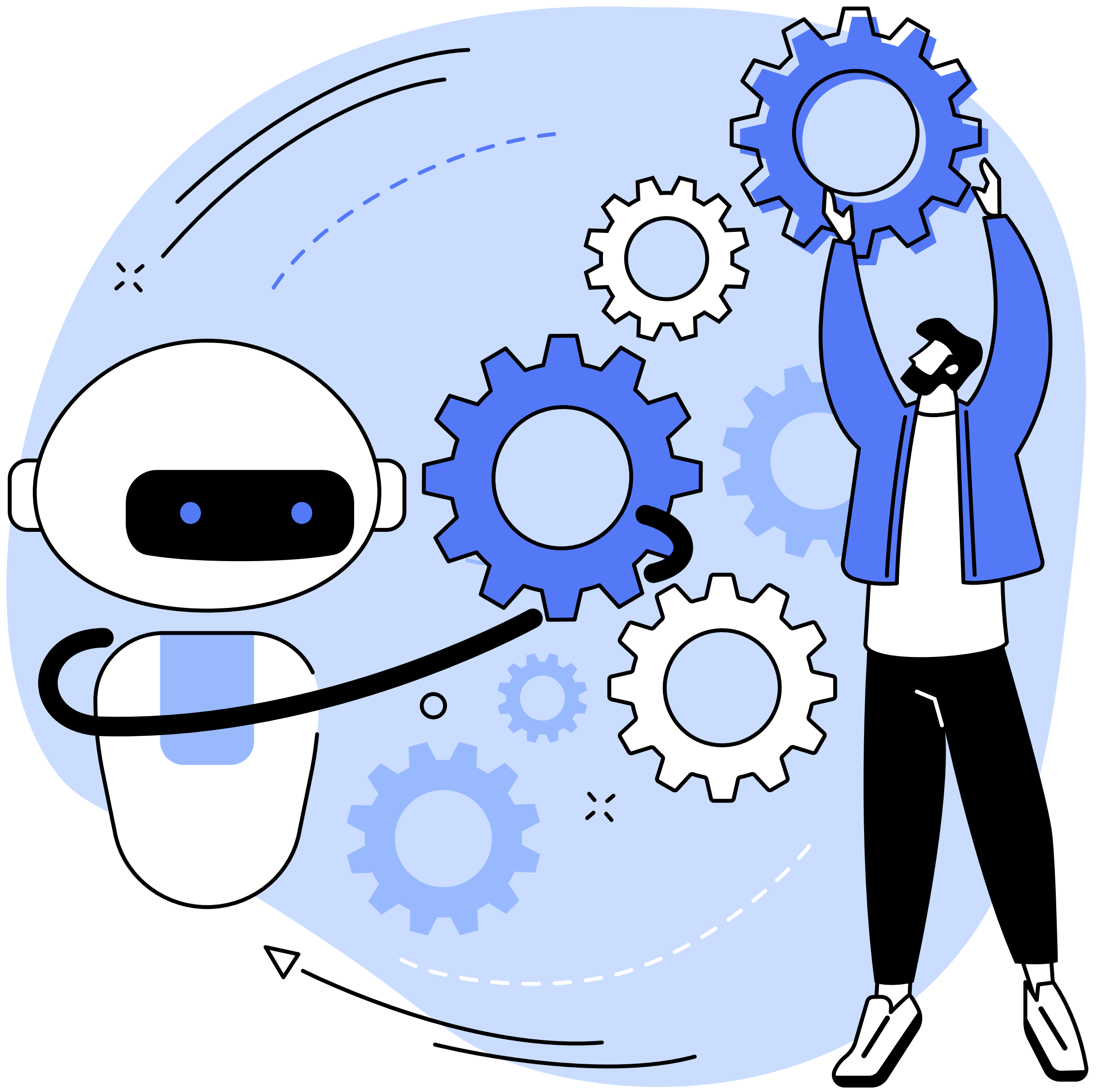
1. **Functional:** plt.plot() – quick and simple
2. **Object-Oriented:** fig, ax = plt.subplots() – better control

- **Chart types created:**
- Line, Bar, Histogram, Scatter, Boxplot

- **Customizations:**
- Titles, axes labels, grid, color, legend, saving figures



EXPLORATORY DATA ANALYSIS (EDA)



- **Objective:** Summarize dataset’s main characteristics

- **Steps followed:**
- Univariate analysis (distributions)
- Bivariate/multivariate analysis (relationships)
- Detect missing values, outliers, class imbalance

EDA – TOOLS AND EXAMPLES

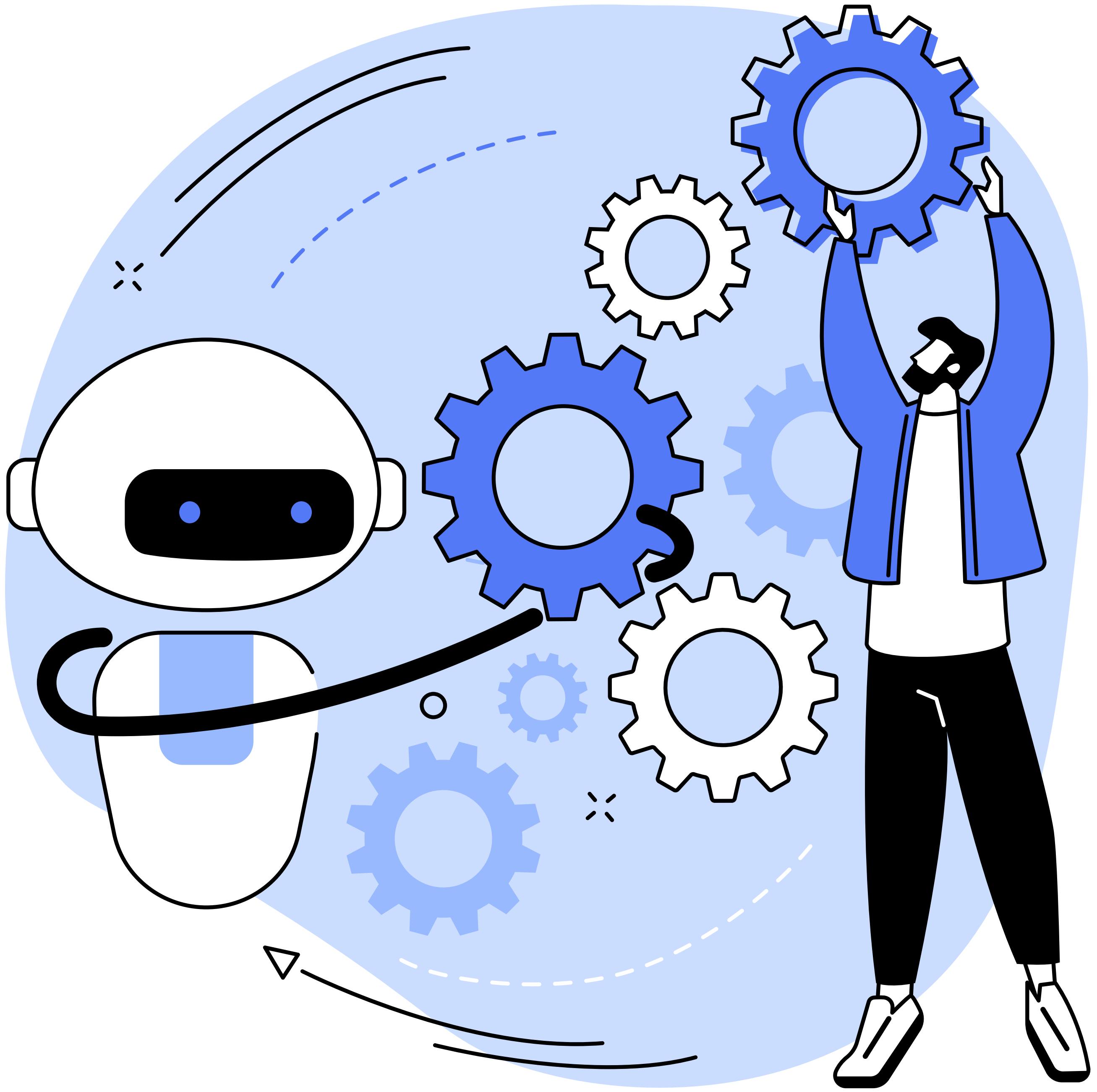
- **df.describe(), df.corr()** for numerical stats
- **Plots used:**
  - Histogram, KDE plots for distributions
  - Box plots for outliers
  - Pairplots and heatmaps for correlation
- Learned to interpret visual trends for early hypotheses

SUMMARY & REFLECTIONS

- Learned to clean and analyze data using Python
- Developed beginner-friendly visualizations
- Understood how to extract patterns from data
- Ready for next phase: Feature engineering and modeling
- Appreciate importance of data quality before ML

EXAMPLES OF ARTIFICIAL INTELLIGENCE

- Chatbots
- Smart assistants
- E-Payments
- Search algorithms
- Media streaming



- Smart cars
- Navigation apps
- Facial recognition
- Text editors
- Social media feeds

PROBLEMS AI CAN POSE TO EDUCATION

Lack of Personalisation

Although AI tools are constantly evolving, they lack the human emotions that are sometimes necessary for effective teaching and learning.

Bias and Discrimination

Bias can occur in AI systems because of the data used to train them. AI algorithms may reflect societal biases, prejudices, and stereotypes.

Dependence on Technology

Both students and teachers may become too dependent on technology, hindering their ability to think critically or solve problems.

Data Privacy and Security

AI-powered tools in general often collect vast amounts of personal data, which might pose a threat to students' privacy and security.

IDEAS TO HELP STUDENTS USE AI TOOLS EFFECTIVELY

Introduce AI tools

Adopt AI tools that can help students with their language learning: translation software, speech recognition software, or writing tools.

Provide support

Teach students these tools and offer tips or step-by-step instructions to use them more efficiently. Monitor their work and give feedback.

Encourage critical thinking

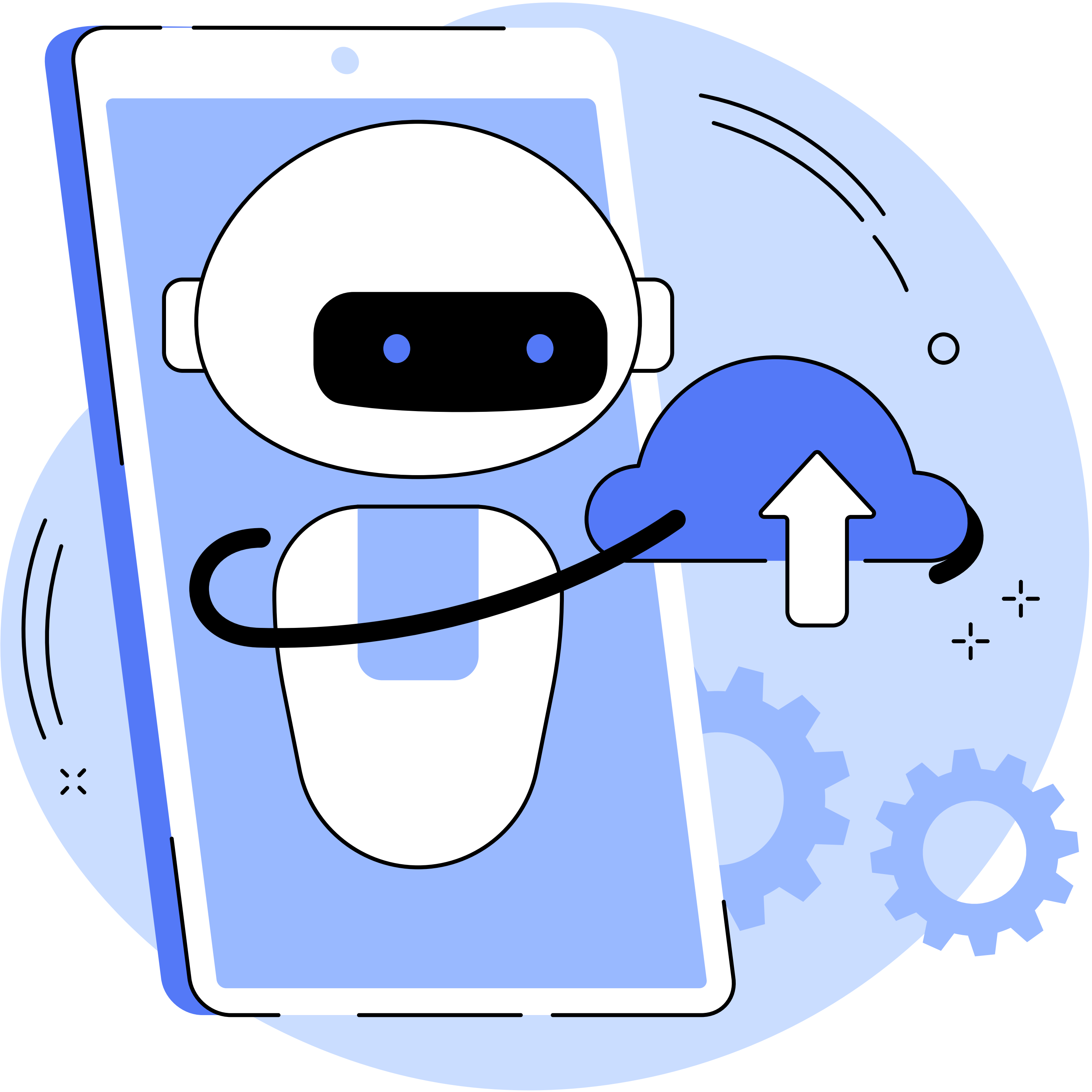
Encourage students to question the accuracy and fairness of AI-powered writing tools or translation apps, and focus on the context.

Keep up-to-date

Teachers need to have their fingers on the pulse and identify new tools and resources that can benefit their students.

HOW WOULD YOU USE AI IN CLASS?

Do you know any **AI-powered tools** that can be useful in the EFL classroom? Let's brainstorm suggestions on the following slide.



THANK YOU FOR LISTENING!